New Siblands School

Mathematics – Shape, Space and Measure Curriculum

Learning bands 1–3: Belonging (Pre-formal Maths) **Mathematics - Shape, Space and Measure**

Band	Spatial Awareness	Shape	Pattern and Statistics	Measures
1	• I can explore a space when I am free to move, roll and stretch. • I can develop an awareness of my body, that my body has different parts and where these are in relation to each other.	• I can explore differently sized and shaped objects. • I can begin to put objects of similar shape inside others and take them out again.	• I can show interest in patterned songs and rhymes (e.g. repeated actions). • I can experience patterned objects and images. • I can begin to predict what happens next in predictable situations.	**Size:** • I can respond to very big or very small items.
2	• I can follow objects or sounds as they are being counted or moved. • I can explore space around me and engage with position and direction (e.g. through pointing, moving or placing objects)	• I can stack objects using flat surfaces. • I can attempt to match shapes with spaces on inset puzzles.	• I can join in with repeated patterns in songs and stories. • I can initiate and continue repeated actions (e.g. resonance board, instruments).	**Size:** • I can explore different sizes in play. **Capacity:** • I can empty containers. **Time:** • I can show awareness of routines.
3	• I can engage in filling and emptying a container. • I can move objects into and out of spaces (e.g. putting objects in and out of boxes). • I can explore objects that are near and far away.	• I can push objects through different shaped holes (e.g. postbox). • I can attempt to fit shapes into spaces on inset boards or puzzles. • I can begin to select a shape for a specific space (e.g. shape sorters). • I can enjoy using blocks to create my own simple structures and arrangements. • I can match objects with parts that fit together (e.g. putting the lid on a teapot).	• I can recognise patterns in my daily routines. • I can anticipate what comes next in a story or rhyme. • I can begin to arrange items in my own patterns (e.g. lining up toys). • I can sort by colour with some help.	**Size/Weight:** • I can explore size and weight. **Capacity:** • I can fill and empty containers (e.g. toys in a pram). **Time:** • I can begin to understand now/later.

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4	• I can begin to remember my way around familiar environments. • I can respond to some spatial and positional language in play (e.g. building train tracks, small world play). • I can respond to simple positional language such as 'in', 'on', 'under'.	• I can choose puzzle pieces and try to fit them in. • I can respond to some spatial and positional language in play (e.g. building train tracks, small world play). • I can make simple constructions.	• I can anticipate repeated sound and action patterns.	**All:** • I can respond to differences in size, weight, length and capacity. **Size:** • I can order items by size (e.g. stacking cups). **Time:** • I can understand time language (before, next, later).
5	• I can respond to and use language of position and direction. • I can predict, move and rotate objects to fit space or create a shape that I would like (e.g. block play to build bridges and enclosures, building tracks, jigsaw puzzles). • I can solve simple problems involving position and movement.	• I can choose items based on shape which are appropriate for their purpose (e.g. triangle block for roof). • I can respond to informal language about shapes (e.g. pointy, twisty, bumpy, wiggly). • I can respond to common shape names (e.g. circle, square, triangle). • I can partition and combine shapes to make new shapes (e.g. cutting food in different ways, construction toys). • I can attempt to create arches and enclosures when building using trial and improvement to select blocks.	• I can create my own patterns. • I can explore simple repeating patterns (AB, ABC). • I can join in with movement and sound patterns.	**All:** • I can compare two items (longer/shorter, heavier/lighter). **Time:** • I can sequence events (daily routines, stories).

6	• I can use spatial language including following and giving directions. • I can follow multi-step directions involving position and movement (e.g. “go under the table then behind the chair”). • I can begin to understand and use ‘left’ and ‘right’ in familiar contexts. • I can describe where objects are in relation to each other without adult support. • I can use position or direction language during play or in the outdoors to explore sequences of body movements (e.g. obstacle courses, directing a friend). • I can use these skills in different contexts (e.g. classroom, outdoors, real-life situations).	• I can use mathematical terms to describe shapes (e.g. straight, curved). • I can combine shapes to make other shapes (e.g. triangles to make rectangles, tangram puzzles). • I can make models of increasing complexity selecting blocks needed, solving problems, and visualising what I will build. • I can sort shapes using simple properties such as curved and straight sides. • I can recognise when shapes or patterns look the same on both sides (early symmetry). • I can solve problems by selecting appropriate shapes.	• I can spot patterns in the environment. • I can identify simple pattern rules. • I can create repeating patterns beyond AB.	**All:** • I can use measuring tools (e.g. rulers, scales). • I can measure using non-standard units (e.g. cubes, hands). **Time:** • I can compare durations (longer/shorter). • I can sequence events.
7	• I understand how moving myself allows objects or images to be seen from different perspectives. • I can visualise how things will appear when turned around. • I can notice and communicate the results of rotating and reflecting images. • I can use the language of position and direction (e.g. in, on, under, up, down, across). • I can use terms which are relative to viewpoint (e.g. in front of, behind, forwards, backwards).	• I can explore and select shapes to fulfil a need in construction and play. • I can discuss how models are built and why shapes were chosen. • I can represent objects using shapes (e.g. circle for a wheel). • I can explain my choices when constructing. • I can show intentional selection (e.g. cylinders because they roll). • I can recognise shapes in different orientations. • I can use shapes to solve practical problems in construction or play.	• I can make patterns (e.g. construction, printing or drawing) and identify shapes used. • I can identify shapes within other shapes. (e.g. small triangles making a bigger triangle or identifying 2D faces of 3D shape). • I can predict what happens when shapes are cut or combined (e.g. cutting/folding paper). • I can sort objects into groups and say how many are in each group. • I can compare groups using language such as more, less or the same.	**All:** • I can use measuring tools consistently. • I can begin to tackle problems involving measure. • I can solve practical problems (e.g. which container holds more). • I can compare measurements when solving problems. **Time:** • I can sequence events accurately.

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8	• I can make constructions, patterns and pictures, and select shapes which will fit when rotated or flipped (e.g. insert boards, shape sorters, jigsaws). • I can use viewpoint language during small world play and model building. • I can draw representations of spatial relationships (e.g. simple maps of a model from above). • I can describe position, direction and movement using mathematical language (e.g. whole, half, quarter turns). • I can use these skills in practical or real-life situations.	• I can recognise and name common 2-D and 3-D shapes, including: rectangles (including squares), circles, triangles. cuboids (including cubes), pyramids and spheres • I can describe properties (e.g. number of sides, edges, faces, corners). • I can use shape properties in construction (e.g. selecting equal-length blocks to make a rectangle).	• I can use a range of shapes, objects or numbers to create patterns and describe what I have made using mathematical language. • I can recognise and describe how a pattern repeats. • I can continue a repeating pattern and explain what I have done. • I can discuss how patterns are built and why certain elements are used. • I can represent groups using objects, marks, pictures. • I can organise groups in a clear way to help me represent data (e.g. lining up, grouping or sorting objects).	**All:** • I can compare, describe and solve practical problems for: lengths and heights (e.g. long/short, longer/shorter, tall/short, double/half) mass/weight (e.g. heavy/light, heavier, lighter) capacity and volume (e.g. full/empty, more/less, half, quarter) • I can measure and record using standard units. **Time:** • I can tell time to the hour and half past. • I can use date-related language. **Money:** • I can recognise coins and notes.
9	• I can order and arrange combinations of mathematical objects in patterns and sequences. • I can use mathematical vocabulary to describe position, direction and movement (e.g. straight line movement, clockwise and anticlockwise turns).	• I can identify and describe the properties of 2-D shapes, including number of sides and line symmetry • I can identify and describe the properties of 3-D shapes, including the number of edges, vertices and faces. • I can identify 2-D shapes on the surface of 3-D shapes (e.g. a circle on a cylinder). • I can compare and sort shapes and everyday objects. • I can discriminate between similar shapes (e.g. different triangles).	• I can interpret and construct pictograms, tally charts, block diagrams and tables. • I can ask and answer simple questions by counting the number of objects in each category and sorting the categories by quantity. • I can ask and answer questions about totalling and comparing categorical data • I can use objects or pictures to help me answer questions about groups (e.g. using counters or drawings to represent data).	**All:** • I can measure using standard units (m/cm, kg/g, l/ml) and compare using >, < and =. **Time:** • I can tell and write the time to five minutes, including quarter past and quarter to. • I can know the number of minutes in an hour and hours in a day. **Money:** • I can solve simple money problems (e.g. combine amounts, finding different combinations.)

10	• I am starting to recognise angles as a property of shape or a description of a turn. • I am starting to identify right angles and relationships between turns. • I am starting to identify whether angles are greater or less than a right angle. • I am starting to identify horizontal, vertical, perpendicular and parallel lines.	• I am starting to draw 2-D shapes. • I am starting to make 3-D shapes using modelling materials. • I am starting to recognise shapes in different orientations and describe them.	• I am starting to interpret and present data using charts, pictograms and tables • I am starting to solve one-step and two-step questions (e.g. 'How many more?' and 'How many fewer?') using information presented in scaled charts, pictograms and tables.	**All:** • I am starting to measure, compare, add and subtract measures. **Size:** • I am starting to measure the perimeter of simple 2-D shapes. **Time:** • I am starting to tell and write the time using analogue clocks (including Roman numerals and 12/24 hour time). • I am starting to estimate and read time to the nearest minute. • I am starting to compare durations and use units of time (seconds, minutes, hours). **Money:** • I am starting to solve problems involving adding and subtracting money, including giving change